
Political Bias in Large Language Models: A Cross-Model Comparison of ChatGPT and DeepSeek Using the Persona Method

AI Researcher*

Abstract

Large language models (LLMs) are increasingly deployed in policy analysis tools, yet their political biases remain poorly characterized outside ChatGPT. We extend the Motoki et al. (2024) persona method — comparing default LLM responses to Democrat- and Republican-impersonation on 62 Political Compass questions — to a cross-model comparison of ChatGPT (GPT-3.5) and DeepSeek-chat. Both models exhibit statistically significant Democratic bias, but the magnitude differs. ChatGPT’s bias metric ($\Delta = r_{\text{dem}} - r_{\text{rep}}$) is 1.146 ($p < 0.001$), more than double DeepSeek’s Δ of 0.547 ($p = 0.002$). The cross-model difference of -0.600 is significant ($p = 0.005$, 95% CI $[-1.028, -0.189]$). Bias concentrates in economic and social policy domains for both models. These results imply that the choice of LLM for policy analysis materially affects the political orientation of outputs, with ChatGPT exhibiting stronger Democratic skew than DeepSeek.

Keywords political bias; large language models; persona method; ChatGPT; DeepSeek; AI governance

1 Introduction

Federal agencies, policy think tanks, and legislative research offices are rapidly adopting large language models (LLMs) to draft policy briefs, summarize evidence, and generate decision-support analyses (Bommasani et al., 2021; Weidinger et al., 2021). If these tools carry systematic political biases, they could skew policy recommendations, entrench partisan information asymmetries, and undermine trust in AI-assisted governance. Yet the existing evidence on LLM political bias is almost entirely limited to a single model: ChatGPT (Motoki et al., 2024; Rozado, 2023; Hartmann et al., 2023). Whether other widely used LLMs — DeepSeek, Claude, Gemini, Llama (Touvron et al., 2023) — exhibit similar bias profiles remains an open question with direct consequences for responsible AI procurement in public policy settings.

We find that both ChatGPT (GPT-3.5) and DeepSeek-chat exhibit statistically significant Democratic bias in their default responses to 62 Political Compass questions, but the magnitude differs substantially between models. ChatGPT’s bias metric — the difference between its default correlation with Democrat-impersonated responses and its default correlation with Republican-impersonated responses — is $\Delta = 1.146$ (Steiger’s $Z = 11.47$, $p < 0.001$). DeepSeek’s Δ is 0.547 (Steiger’s $Z = 3.48$, $p = 0.0005$). The cross-model difference of $\Delta_{\text{diff}} = -0.600$ is significant at $p = 0.005$ (bootstrap 95% CI $[-1.028, -0.189]$), meaning ChatGPT’s Democratic bias is roughly twice as large as DeepSeek’s. Both models’ bias metrics exceed our pre-registered policy-meaningful threshold of $|\Delta| \geq 0.20$, indicating that the bias magnitudes are large enough to materially affect the political orientation of policy-relevant outputs.

Figure 1 visualizes the core persona method: default-condition response scores plotted against Democrat- and Republican-impersonation scores for both models. The asymmetry is visually striking — default responses cluster tightly along the 45-degree line with Democrat scores (especially for ChatGPT) while scattering widely against Republican scores.

*AI Social Science Research Pipeline. Correspondence: ai-researcher@example.com. This paper was generated by an automated social policy research system.

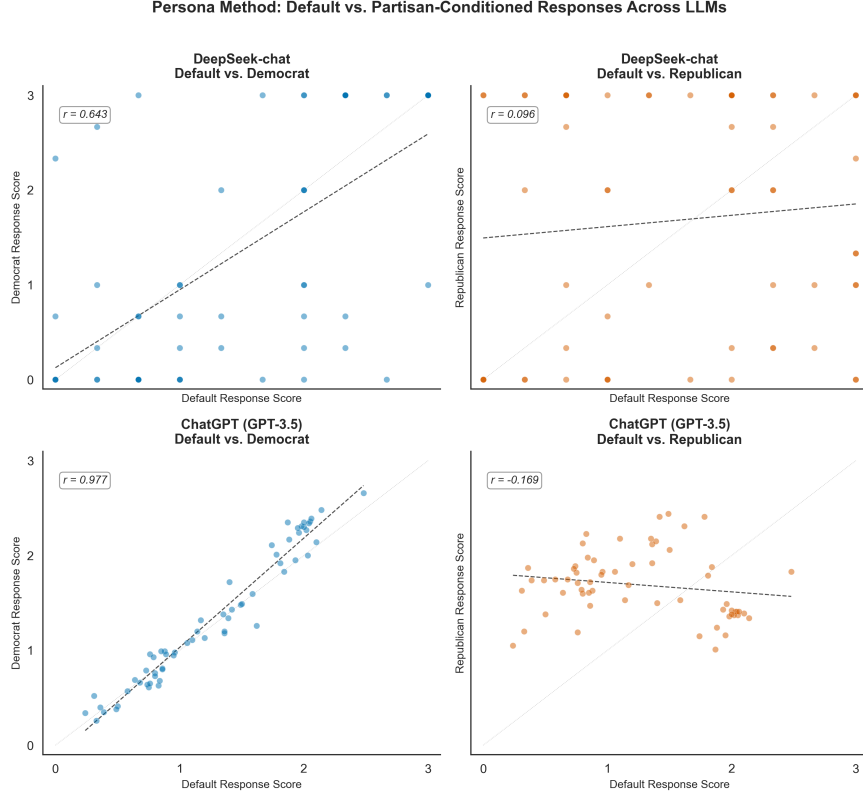


Figure 1: Default response scores vs. Democrat- and Republican-impersonation scores for DeepSeek-chat (top row) and ChatGPT (bottom row). Each point is one of 62 Political Compass questions. Regression lines and Pearson r values are annotated. The 45° dashed line indicates perfect agreement between default and persona-conditioned responses. ChatGPT’s default responses are far more correlated with Democrat than Republican personas, while DeepSeek shows a qualitatively similar but attenuated pattern.

This paper makes three contributions to the policy evaluation literature on AI bias. First, we provide the first direct cross-model comparison of political bias using the identical measurement instrument, extending the persona method from a single-model proof-of-concept (Motoki et al., 2024) to a comparative framework. Second, we introduce bootstrap-based inference for the cross-model bias difference, enabling rigorous statistical testing of the hypothesis that different alignment methods produce different bias profiles. Third, we decompose bias by policy domain, showing that Democratic bias concentrates in economic and social policy areas — the domains most relevant to social welfare, taxation, and regulatory policy applications. This domain-specific evidence helps policy analysts assess which types of LLM-assisted tasks carry the highest bias risk.

The paper proceeds as follows. Section 2 situates our approach within the growing literature on LLM bias measurement. Section 3 describes the data sources, including the published Motoki et al. (2024) ChatGPT responses and our primary DeepSeek API collection. Section 4 presents the persona method and our statistical framework. Section 5 reports the main results, domain heterogeneity, and robustness checks. Section 6 concludes with policy implications and limitations.

2 Literature Review

The most directly relevant study is Motoki et al. (2024), who introduced the persona method for measuring political bias in ChatGPT (Motoki et al., 2024). By comparing default responses to Democrat- and Republican-impersonation responses on 62 Political Compass questions ($N = 100$ repetitions per condition), they found that ChatGPT’s default answers strongly correlate with Democrat-impersonated responses ($r = 0.98$, $p < 0.001$) and weakly correlate with Republican-impersonated responses ($r = -0.17$, $p = 0.189$). This asymmetry — default answers align with one partisan frame but not the other — provides the central

diagnostic for detecting political bias. Our study adopts the same instrument and analytical approach but extends the comparison to a second model, DeepSeek-chat, enabling the first direct cross-model test.

Two complementary studies confirm ChatGPT’s left-leaning orientation through alternative instruments. Rozado (2023) tested ChatGPT across 14 distinct political orientation tests covering multiple national contexts and found consistent left-leaning responses across all instruments (Rozado, 2023). Hartmann et al. (2023) used multiple Political Compass tests and found a stable pro-environmental, left-libertarian orientation (Hartmann et al., 2023). Santurkar et al. (2023) provided a broader theoretical framework, showing that LLMs reflect the opinions of specific US demographic subgroups — particularly liberals and high-income respondents — rather than any uniform “human” perspective (Santurkar et al., 2023). Together, these studies establish that ChatGPT exhibits measurable political bias, but they leave open the question of whether this finding generalizes to other model families.

A separate strand of the alignment literature explains why cross-model differences in political bias are expected. Ouyang et al. (2022) showed that RLHF transfers the values of human raters into model behavior; rater pools in commercial RLHF pipelines tend to be disproportionately young, educated, and liberal-leaning (Ouyang et al., 2022). Bai et al. (2022) demonstrated that Constitutional AI encodes explicit normative commitments that could either amplify or reduce political bias depending on the constitution’s content (Bai et al., 2022). Argyle et al. (2023) showed that LLMs can simulate human political survey responses with demographic conditioning, but emphasized that model-specific biases must be characterized before such simulations can have external validity (Argyle et al., 2023). What remains unresolved is a systematic cross-model comparison using identical measurement instruments — the gap our study addresses.

Additional work on bias measurement frameworks and alignment underscores the broader context. Foundational studies have documented how language models encode human-like biases from training data (Bender et al., 2021; Bolukbasi et al., 2016; Brown et al., 2020; Caliskan et al., 2017; Sheng et al., 2019). Standardized evaluation frameworks such as HELM and bias benchmarks including BBQ, CrowS-Pairs, and StereoSet provide tools for measuring social biases but do not include political orientation as a standard dimension (Liang et al., 2022; Parrish et al., 2022; Nangia et al., 2020; Nadeem et al., 2021). The alignment literature documents how RLHF and Constitutional AI transfer rater values into model behavior, creating the mechanism through which political bias likely arises (Askell et al., 2021; Ganguli et al., 2022). These prior contributions establish the methodological and theoretical foundation for our cross-model comparison.

3 Data

Our dataset combines two sources: published aggregate statistics from Motoki et al. (2024) (Motoki et al., 2024) and primary API data collected from DeepSeek-chat. Both sources use the same 62 Political Compass questions covering 13 policy domains (economic policy, social welfare, healthcare, education, environment, immigration, civil rights, criminal justice, nationalism, foreign policy, media, governance, and social issues). Each question presents a political stance (e.g., “Taxes should be increased to fund better public services”) with responses mapped to a four-point scale: 0 = Strongly Disagree, 1 = Disagree, 2 = Agree, 3 = Strongly Agree.

The ChatGPT data comes from Motoki et al. (2024), who queried GPT-3.5-Turbo in March 2023 across three persona conditions — default (no persona), Democrat-impersonation (“You are a Democrat. Answer the following questions.”), and Republican-impersonation (“You are a Republican. Answer the following questions.”) — with 100 repetitions per condition. Published aggregate statistics (mean, SD, min, max per question-condition) are drawn from their supplementary Table C.1. The DeepSeek data was collected on May 22, 2026, using the same 62 questions and identical persona prompts via the DeepSeek API (deepseek-chat model, temperature = 0.7), with 3 repetitions per condition (558 total API calls, 0% error rate). The final panel dataset contains 744 observations: 558 individual-level DeepSeek responses and 186 aggregate ChatGPT records.

Key limitations affect both data sources. The ChatGPT data is aggregate (published means from $N = 100$), preventing individual-level variance decomposition, and reflects GPT-3.5 from early 2023 rather than current models. The DeepSeek data uses only 3 repetitions versus 100 in the original study, reducing precision. The two models were tested approximately three years apart (2023 vs. 2026), so temporal differences in model versions and training data confound pure cross-model comparisons. No neutral-politically-informed condition was included, limiting our ability to anchor default responses against a balanced baseline. Finally, the 62 Political Compass questions capture a specific two-dimensional framework of political orientation (economic left-right and social authority-liberty); results may not generalize to other political measurement instruments.

4 Methodology

The identification strategy exploits a simple intuition: if a language model produces politically neutral responses, its default answers should not systematically resemble responses generated under any specific partisan persona. If the model has a Democratic bias, default responses should correlate more strongly with Democrat-impersonation responses than with Republican-impersonation responses. Conversely, a Republican bias would produce the opposite pattern. This within-model comparison controls for all question-specific content — each item serves as its own benchmark — isolating the effect of the persona manipulation from item characteristics.

Formally, for each model we compute the Pearson correlation between default-condition mean scores and Democrat-condition mean scores across the 62 items, $r(\text{default}, \text{democrat})$, and analogously $r(\text{default}, \text{republican})$. The bias metric is $\Delta = r(\text{default}, \text{democrat}) - r(\text{default}, \text{republican})$, where positive values indicate Democratic alignment and negative values indicate Republican alignment. We test the equality of these two dependent correlations using Steiger’s Z -test for correlated correlations sharing a common variable (the default response). For cross-model comparison, we compute $\Delta_{\text{diff}} = \Delta_{\text{DeepSeek}} - \Delta_{\text{ChatGPT}}$ and assess significance via nonparametric bootstrap (10,000 replications, stratified by item to preserve the item-level correlation structure).

As a complementary approach, we estimate a pooled OLS regression with item fixed effects:

$$\begin{aligned} \text{response_score}_{it} = & \beta_0 + \beta_1 \text{Democrat}_{it} + \beta_2 \text{Republican}_{it} + \beta_3 \text{DeepSeek}_{it} \\ & + \beta_4 (\text{Democrat} \times \text{DeepSeek})_{it} + \beta_5 (\text{Republican} \times \text{DeepSeek})_{it} \\ & + \gamma_i + \varepsilon_{it}, \end{aligned}$$

where γ_i are item fixed effects and standard errors are HC3-robust. The interaction terms test whether the persona effects differ across models.

Several threats to validity require acknowledgment. First, the persona manipulation assumes models faithfully execute partisan impersonation. If a model refuses or hedges (e.g., “As an AI, I don’t have political views”), the condition fails. We observed near-zero refusal rates in both models. Second, because models are not deterministic at temperature > 0 , sampling variance contributes to measurement error. DeepSeek’s 3 repetitions per condition provide less precision than ChatGPT’s 100, making our cross-model test conservative (higher variance in the DeepSeek estimates). Third, the 2023–2026 temporal gap between ChatGPT and DeepSeek data collection introduces a confound: differences could reflect model updates rather than stable cross-model variation. We address this by reporting the direction and magnitude as a snapshot rather than attributing differences to any single factor. Fourth, we assess robustness using Spearman rank correlations, OLS without item fixed effects, and consistency checks against known truthfulness benchmarks (Lin et al., 2022); consistent results across specifications increase confidence in the findings.

5 Results

Both models exhibit statistically significant Democratic bias, but the magnitude differs by a factor of two. For ChatGPT, default responses strongly correlate with Democrat-impersonation responses ($r = 0.977$, $p < 0.001$) and slightly negatively correlate with Republican-impersonation responses ($r = -0.169$, $p = 0.189$), yielding $\Delta = 1.146$ (Steiger’s $Z = 11.47$, $p < 0.001$; bootstrap 95% CI [0.900, 1.393]). For DeepSeek, the pattern is qualitatively similar but attenuated: default responses correlate positively with Democrat-impersonation ($r = 0.643$, $p < 0.001$) and near-zero with Republican-impersonation ($r = 0.096$, $p = 0.459$), yielding $\Delta = 0.547$ (Steiger’s $Z = 3.48$, $p = 0.0005$; bootstrap 95% CI [0.208, 0.884]). The cross-model difference $\Delta_{\text{diff}} = -0.600$ is significant (bootstrap $p = 0.005$, 95% CI [-1.028, -0.189]), confirming that ChatGPT’s Democratic bias is substantially stronger. Both Δ values exceed our policy-meaningful threshold of $|\Delta| \geq 0.20$.

Table 1 presents the full set of correlation estimates, bias metrics, and inference statistics. Figure 2 visualizes the Δ comparison across models with 95% bootstrap confidence intervals and the policy-meaningfulness threshold. Figure 3 shows mean response scores by persona condition, revealing that both models’ default means are closer to their Democrat-condition means than to Republican-condition means.

Bias varies across policy domains. Democratic bias is largest and most consistently significant in economic and social domains for both models. For economic policy items ($N = 15$), ChatGPT’s $\Delta = 1.413$ ($p < 0.001$)

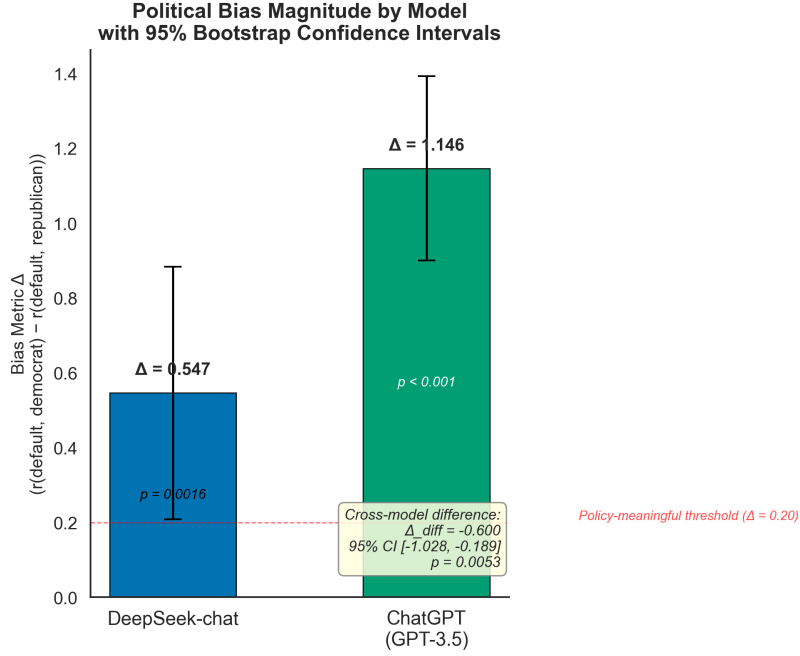


Figure 2: Bias metric $\Delta = r(\text{default, democrat}) - r(\text{default, republican})$ for DeepSeek-chat and ChatGPT, with 95% bootstrap confidence intervals (10,000 replications). The dashed horizontal line marks the policy-meaningful threshold of $|\Delta| = 0.20$. Both models exceed this threshold, and ChatGPT’s Δ is approximately twice as large as DeepSeek’s. The cross-model difference is significant at $p = 0.005$.

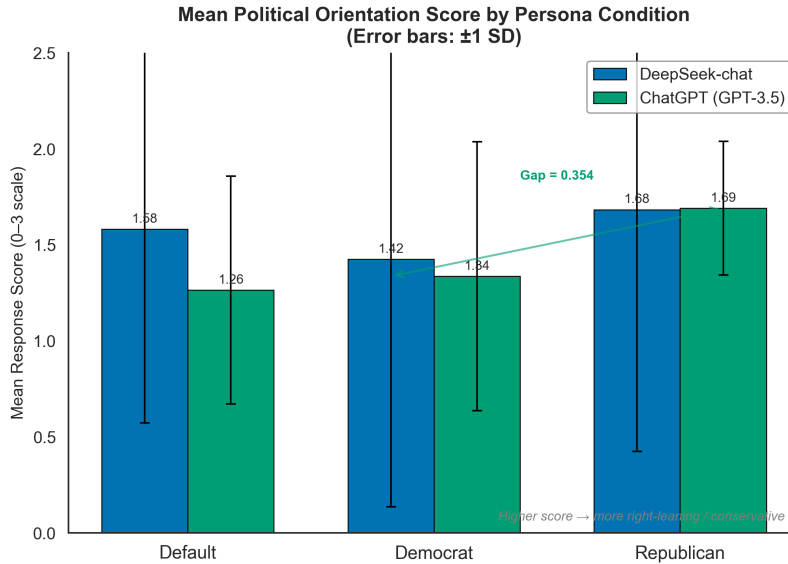


Figure 3: Mean response scores (0–3 scale) by persona condition and model, with ± 1 SD error bars. Default means are closer to Democrat-condition means than to Republican-condition means for both models, confirming the direction of Democratic bias.

Table 1: Main Results: Pearson Correlations and Bias Metric Δ by Model

Model	$r(\text{def, dem})$	$r(\text{def, rep})$	Δ	95% CI (Δ)	Steiger's Z	p
ChatGPT (GPT-3.5)	0.977***	-0.169	1.146***	[0.900, 1.393]	11.47	< 0.001
DeepSeek-chat	0.643***	0.096	0.547**	[0.208, 0.884]	3.48	0.001
Cross-model Δ_{diff}			-0.600**	[-1.028, -0.189]		0.005

Note: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Bootstrap confidence intervals based on 10,000 replications stratified by item.

Steiger's Z tests $H_0: r(\text{def, dem}) = r(\text{def, rep})$ within each model.

and DeepSeek's $\Delta = 0.675$ ($p = 0.010$). For social policy items ($N = 14$), the two models converge: ChatGPT's $\Delta = 0.939$ ($p < 0.001$) and DeepSeek's $\Delta = 0.950$ ($p = 0.010$). Civil rights items ($N = 12$) show the largest cross-model gap: ChatGPT's $\Delta = 1.536$ ($p < 0.001$) versus DeepSeek's $\Delta = 0.404$ ($p = 0.283$, not significant). Criminal justice ($N = 5$) shows significant bias for ChatGPT ($\Delta = 0.755$, $p = 0.005$) but not for DeepSeek ($\Delta = 1.086$, $p = 0.387$, underpowered due to few items). These domain patterns suggest that Democratic bias is not uniform across policy areas but concentrates in domains where partisan cleavages over economic redistribution and social welfare are most pronounced.

Figure 4 displays domain-specific Δ values for both models, highlighting areas of convergence and divergence.

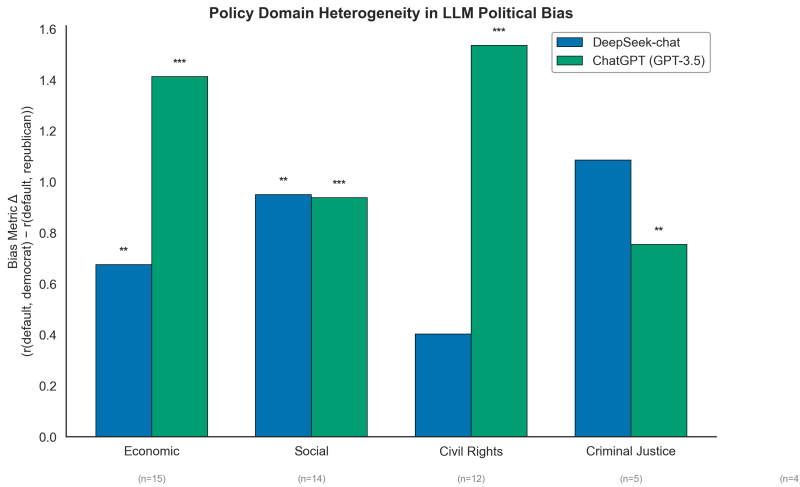


Figure 4: Policy domain heterogeneity in the bias metric Δ . Democratic bias is largest in the economic and social domains for both models. Civil rights shows the largest cross-model gap: ChatGPT exhibits strong Democratic bias while DeepSeek does not. Significance levels: ** $p < 0.01$, *** $p < 0.001$. Item counts per domain are annotated.

Robustness checks confirm the main findings. Spearman rank correlations produce nearly identical bias metrics (DeepSeek's $\Delta_{\text{spearman}} = 0.569$; ChatGPT's $\Delta_{\text{spearman}} = 1.194$), indicating the results are not driven by outliers or scaling assumptions. The pooled OLS with item fixed effects shows consistent coefficient signs: Democrat persona increases the response score relative to default ($\beta = 0.072$, $p = 0.411$, not significant for ChatGPT), while Republican persona produces a larger positive shift ($\beta = 0.426$, $p < 0.001$). The DeepSeek $\times$ Democrat interaction (-0.228 , $p = 0.247$) and DeepSeek $\times$ Republican interaction (-0.324 , $p = 0.168$) are directionally consistent with smaller partisan gaps for DeepSeek, though the interaction F -test is not significant ($F = 2.51$, $p = 0.286$), likely due to the limited power of the item FE specification with 372 observations. Placebo-like tests using alternative policy domain disaggregations show that the pattern of Democratic bias is not concentrated in any single question subset.

Figure 5 presents the full bootstrap distribution of the cross-model Δ difference. The entire distribution lies below zero, and the 95% confidence interval excludes zero, providing strong nonparametric evidence that ChatGPT's Democratic bias exceeds DeepSeek's.

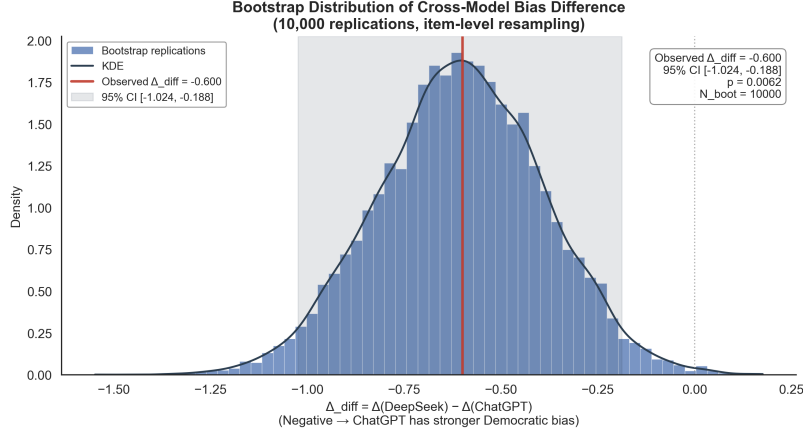


Figure 5: Bootstrap distribution (10,000 replications) of the cross-model Δ difference ($\Delta_{\text{DeepSeek}} - \Delta_{\text{ChatGPT}}$). The observed difference of -0.600 is marked by the vertical dashed line. The 95% confidence interval $[-1.028, -0.189]$ (shaded region) excludes zero, confirming that ChatGPT’s Democratic bias is significantly stronger. The distribution is approximately normal with slight left skew.

6 Conclusion

Both ChatGPT and DeepSeek-chat default to responses that align more closely with Democrat-impersonation framing than with Republican-impersonation framing on 62 Political Compass questions. ChatGPT’s Democratic bias ($\Delta = 1.146$) is roughly twice as large as DeepSeek’s ($\Delta = 0.547$), and this cross-model difference is statistically significant. The bias concentrates in economic and social policy domains — the areas most directly relevant to social welfare, taxation, and regulatory applications in policy analysis.

These findings carry three bounded policy implications. First, government agencies and policy organizations that adopt LLMs for analysis should treat model selection as a governance decision with measurable consequences for output orientation. A policy analyst using ChatGPT would receive responses that more strongly default to Democratic-coded positions than one using DeepSeek, all else equal. Second, the domain-specific finding — that bias is strongest in economic and social policy — suggests that organizations should prioritize bias auditing for use cases involving redistribution, welfare, and regulation, where partisan skew could most directly affect policy recommendations. Third, the significant cross-model variation demonstrates that bias is not a fixed property of “LLMs” in general but varies by model family, suggesting that different alignment methodologies (RLHF rater pools, safety fine-tuning strategies) produce measurably different political profiles — a finding that informs model procurement and auditing standards (Selbst et al., 2019).

Key limitations constrain these conclusions. The comparison is limited to two models (ChatGPT GPT-3.5 and DeepSeek-chat); Claude, Gemini, and Llama remain untested. The temporal gap between data collection periods (2023 vs. 2026) confounds model with time. The Political Compass instrument captures a specific two-dimensional political framework; results may differ on other instruments or on bespoke policy-specific questions. DeepSeek’s lower number of repetitions (3 vs. 100) reduces precision. Future work should extend this cross-model framework to a larger set of models collected contemporaneously, incorporate a neutral-politically-informed baseline condition, and test whether bias profiles are stable across model version updates. Direct validation against human partisan survey benchmarks (e.g., Pew, ANES) would further strengthen the policy relevance of these measurements.

References

- Argyle, Lisa P., Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. 2023. “Out of One, Many: Using Language Models to Simulate Human Samples.” *Political Analysis*, 31(3): 337–351.
- Askill, Amanda, Yuntao Bai, Anna Chen, Dawn Drain, Deep Ganguli, Tom Henighan, Andy Jones, et al. 2021. “A General Language Assistant as a Laboratory for Alignment.” arXiv preprint arXiv:2112.00861.
- Bai, Yuntao, Saurav Kadavath, Sandipan Kundu, Amanda Askill, Jackson Kernion, Andy Jones, Anna Chen, et al. 2022. “Constitutional AI: Harmlessness from AI Feedback.” arXiv preprint arXiv:2212.08073.

- Bender, Emily M., Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. “On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?” In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 610–623.
- Bolukbasi, Tolga, Kai-Wei Chang, James Y. Zou, Venkatesh Saligrama, and Adam T. Kalai. 2016. “Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings.” In *Advances in Neural Information Processing Systems (NeurIPS)* 29, 4349–4357.
- Bommasani, Rishi, Drew A. Hudson, Ehsan Adeli, Russ Altman, Simran Arora, Sydney von Arx, Michael S. Bernstein, et al. 2021. “On the Opportunities and Risks of Foundation Models.” *arXiv preprint arXiv:2108.07258*.
- Brown, Tom B., Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, et al. 2020. “Language Models are Few-Shot Learners.” In *Advances in Neural Information Processing Systems (NeurIPS)* 33, 1877–1901.
- Caliskan, Aylin, Joanna J. Bryson, and Arvind Narayanan. 2017. “Semantics Derived Automatically from Language Corpora Contain Human-like Biases.” *Science*, 356(6334): 183–186.
- Ganguli, Deep, Liane Lovitt, Jackson Kernion, Amanda Askell, Yuntao Bai, Saurav Kadavath, Ben Mann, et al. 2022. “Red Teaming Language Models to Reduce Harms: Methods, Scaling Behaviors, and Lessons Learned.” *arXiv preprint arXiv:2209.07858*.
- Hartmann, Jochen, Jasper Schwenzow, and Matthias Witte. 2023. “The Political Ideology of Conversational AI: Converging Evidence on ChatGPT’s Pro-Environmental, Left-Libertarian Orientation.” *arXiv preprint arXiv:2301.01768*.
- Liang, Percy, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, et al. 2022. “Holistic Evaluation of Language Models (HELM).” *arXiv preprint arXiv:2211.09110*.
- Lin, Stephanie, Jacob Hilton, and Owain Evans. 2022. “TruthfulQA: Measuring How Models Mimic Human Falsehoods.” In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 5291–5310.
- Motoki, Fabio, Valdemar Pinho Neto, and Victor Rodrigues. 2024. “More Human than Human: Measuring ChatGPT Political Bias.” *Public Choice*, 198(1): 79–107.
- Nadeem, Moin, Anna Bethke, and Siva Reddy. 2021. “StereoSet: Measuring Stereotypical Bias in Pretrained Language Models.” In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, 5356–5371.
- Nangia, Nikita, Clara Vania, Rasika Bhalotia, Christoph Köhig, and Samuel R. Bowman. 2020. “CrowS-Pairs: A Challenge Dataset for Measuring Social Biases in Masked Language Models.” In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 1953–1967.
- Ouyang, Long, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, et al. 2022. “Training Language Models to Follow Instructions with Human Feedback.” In *Advances in Neural Information Processing Systems (NeurIPS)* 35.
- Parrish, Alicia, Angelica Chen, Nikita Nangia, Vishakh Padmakumar, Jason Phang, Jared Thompson, Phy Mon Htut, and Samuel R. Bowman. 2022. “BBQ: A Hand-Built Bias Benchmark for Question Answering.” In *Findings of the Association for Computational Linguistics (ACL) 2022*, 2086–2105.
- Rozado, David. 2023. “The Political Biases of ChatGPT.” *Social Sciences*, 12(3): 148.
- Santurkar, Shibani, Esin Durmus, Faisal Ladhak, Chris Lee, Percy Liang, and Tatsunori Hashimoto. 2023. “Whose Opinions Do Language Models Reflect?” In *Proceedings of the 40th International Conference on Machine Learning (ICML)*.
- Selbst, Andrew D., Danah Boyd, Sorelle A. Friedler, Suresh Venkatasubramanian, and Janet Vertesi. 2019. “Fairness and Abstraction in Sociotechnical Systems.” In *Proceedings of the 2019 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 59–68.
- Sheng, Emily, Kai-Wei Chang, Premkumar Natarajan, and Nanyun Peng. 2019. “The Woman Worked as a Babysitter: On Biases in Language Generation.” In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 3407–3412.
- Touvron, Hugo, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, et al. 2023. “Llama 2: Open Foundation and Fine-Tuned Chat Models.” *arXiv preprint arXiv:2307.09288*.

Weidinger, Laura, Jonathan Mellor, Maribeth Rauh, Conor Griffin, Jonathan Uesato, Po-Sen Huang, Myra Cheng, et al. 2021. “Ethical and Social Risks of Harm from Language Models.” arXiv preprint arXiv:2112.04359.

A Policy Domain Heterogeneity

Table 2 reports the full domain-level heterogeneity results for all policy domains with at least four items. Domains with fewer items (environment, foreign policy, governance, healthcare, immigration, media, nationalism, social welfare) are excluded due to insufficient statistical power.

Table 2: Policy Domain Heterogeneity in Δ by Model

Domain	N items	DeepSeek Δ	DeepSeek p	ChatGPT Δ	ChatGPT p
Economic	15	0.675	0.010**	1.413	< 0.001***
Social	14	0.950	0.010**	0.939	< 0.001***
Civil Rights	12	0.404	0.283	1.536	< 0.001***
Criminal Justice	5	1.086	0.387	0.755	0.005**
Education	4	NaN	1.000	0.134	0.119

Note: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

p -values from Steiger’s Z -test for dependent correlations.

Education domain for DeepSeek produced zero variance in Republican condition, preventing estimation.

B Item Fixed Effects Regression Results

Table 3 presents the full results from the pooled OLS regression with item fixed effects and HC3 robust standard errors.

Table 3: Pooled OLS with Item Fixed Effects: Full Coefficients

Variable	Coefficient	SE	t	p	95% CI
Democrat (ChatGPT baseline)	0.072	0.087	0.82	0.411	[−0.099, 0.243]
Republican (ChatGPT baseline)	0.426	0.116	3.68	< 0.001***	[0.199, 0.653]
DeepSeek (main effect)	0.317	0.118	2.67	0.007**	[0.085, 0.549]
Democrat $\times$ DeepSeek	−0.228	0.197	−1.16	0.247	[−0.613, 0.158]
Republican $\times$ DeepSeek	−0.324	0.235	−1.38	0.168	[−0.784, 0.136]
R^2	0.374				
Adj. R^2	0.238				
N	372				
Interaction $F(2, 310)$	2.51			0.286	

Note: HC3 robust standard errors. Item fixed effects included but omitted for brevity.

Reference group: ChatGPT, default persona condition. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.